

Answer ALL questions. Write your answers in the spaces provided.

1. A researcher is investigating the distribution of orchids in a field. He believes that the Poisson distribution with a mean of 1.75 may be a good model for the number of orchids in each square metre. He randomly selects 150 non-overlapping areas, each of one square metre, and counts the number of orchids present in each square.

The results are recorded in the table below.

Number of orchids in each square metre	0	1	2	3	4	5	6
Number of squares	30	42	35	26	11	6	0

He calculates the **expected** frequencies as follows

Number of orchids in each square metre	0	1	2	3	4	5	More than 5
Number of squares	26.07	45.62	39.91	23.28	10.19	3.57	r

- (a) Find the value of r giving your answer to 2 decimal places.

(1)

The researcher will test, at the 5% level of significance, whether or not the data can be modelled by a Poisson distribution with mean 1.75

- (b) State clearly the hypotheses required to test whether or not this Poisson distribution is a suitable model for these data.

(1)

The test statistic for this test is 2.0 and the number of degrees of freedom to be used is 4

- (c) Explain fully why there are 4 degrees of freedom.

(2)

- (d) Stating your critical value clearly, determine whether or not these data support the researcher's belief.

(2)

The researcher works in another field where the number of orchids in each square metre is known to have a Poisson distribution with mean 1.5

He randomly selects 200 non-overlapping areas, each of one square metre, in this second field, and counts the number of orchids present in each square.

- (e) Using a Poisson approximation, show that the probability that he finds at least one square with exactly 6 orchids in it is 0.506 to 3 decimal places.

(4)



2. The number of heaters, H , bought during one day from *Warmup* supermarket can be modelled by a Poisson distribution with mean 0.7

(a) Calculate $P(H \geq 2)$

(1)

The number of heaters, G , bought during one day from *Pumraw* supermarket can be modelled by a Poisson distribution with mean 3, where G and H are independent.

(b) Show that the probability that a total of fewer than 4 heaters are bought from these two supermarkets in a day is 0.494 to 3 decimal places.

(2)

(c) Calculate the probability that a total of fewer than 4 heaters are bought from these two supermarkets on at least 5 out of 6 randomly chosen days.

(3)

December was particularly cold. Two days in December were selected at random and the total number of heaters bought from these two supermarkets was found to be 14

(d) Test whether or not the mean of the total number of heaters bought from these two supermarkets had increased. Use a 5% level of significance and state your hypotheses clearly.

(5)

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3. Andreia's secretary makes random errors in his work at an average rate of 1.7 errors every 100 words.

- (a) Find the probability that the secretary makes fewer than 2 errors in the next 100-word piece of work. (2)

Andreia asks the secretary to produce a 250-word article for a magazine.

- (b) Find the probability that there are exactly 5 errors in this article. (2)

Andreia offers the secretary a choice of one of two bonus schemes, based on a random sample of 40 pieces of work each consisting of 100 words.

In scheme **A** the secretary will receive the bonus if more than 10 of the 40 pieces of work contain no errors.

In scheme **B** the bonus is awarded if the total number of errors in all 40 pieces of work is fewer than 56

- (c) Showing your calculations clearly, explain which bonus scheme you would advise the secretary to choose. (5)

Following the bonus scheme, Andreia randomly selects a single 500-word piece of work from the secretary to test if there is any evidence that the secretary's rate of errors has decreased.

- (d) Stating your hypotheses clearly and using a 5% level of significance, find the critical region for this test. (4)

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1. A plumbing company receives call-outs during the working day at an average rate of 2.4 per hour.

- (a) Find the probability that the company receives exactly 7 call-outs in a randomly selected 3-hour period of a working day. (2)

The company has enough staff to respond to 28 call-outs in an 8-hour working day.

- (b) Show that the probability that the company receives more than 28 call-outs in a randomly selected 8-hour working day is 0.022 to 3 decimal places. (2)

In a random sample of 100 working days each of 8 hours,

- (c) (i) find the expected number of days that the company receives more than 28 call-outs, (1)
- (ii) find the standard deviation of the number of days that the company receives more than 28 call-outs, (2)
- (iii) use a Poisson approximation to estimate the probability that the company receives more than 28 call-outs on at least 6 of these days. (3)

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4. During the morning, the number of cyclists passing a particular point on a cycle path in a 10-minute interval travelling eastbound can be modelled by a Poisson distribution with mean 8

The number of cyclists passing the same point in a 10-minute interval travelling westbound can be modelled by a Poisson distribution with mean 3

- (a) Suggest a model for the total number of cyclists passing the point on the cycle path in a 10-minute interval, stating a necessary assumption.

(2)

Given that exactly 12 cyclists pass the point in a 10-minute interval,

- (b) find the probability that at least 11 are travelling eastbound.

(3)

After some roadworks were completed, the total number of cyclists passing the point in a randomly selected 20-minute interval one morning is found to be 14

- (c) Test, at the 5% level of significance, whether there is evidence of a decrease in the rate of cyclists passing the point.
State your hypotheses clearly.

(3)

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2. Xena catches fish at random, at a constant rate of 0.6 per hour.

(a) Find the probability that Xena catches exactly 4 fish in a 5-hour period.

(2)

The probability of Xena catching no fish in a period of t hours is less than 0.16

(b) Find the minimum value of t , giving your answer to one decimal place.

(3)

Independently of Xena, Zion catches fish at random with a mean rate of 0.8 per hour.

Xena and Zion try using new bait to catch fish. The number of fish caught in total by Xena and Zion after using the new bait, in a randomly selected 4-hour period, is 12

(c) Use a suitable test to determine, at the 5% level of significance, whether or not there is evidence that the rate at which fish are caught has increased after using the new bait. State your hypotheses clearly and the p -value used in your test.

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- (1)

Number of heads	0	1	2	3	4	5
Observed frequency	2	27	93	181	146	51

Number of heads	0	1	2	3	4	5
Expected frequency	5.12	38.40	115.20	172.80	129.60	38.88

- (b) State the number of degrees of freedom in Chris' test, giving a reason for your answer. (1)
- (c) Carry out the test at the 5% level of significance. (5)
You should state your hypotheses, test statistic, critical value and conclusion clearly.
- (d) Showing your working, find an alternative model which would better fit Chris' data. (2)

3. A machine produces cloth. Faults occur randomly in the cloth at a rate of 0.4 per square metre.

The machine is used to produce tablecloths, each of area A square metres. One of these tablecloths is taken at random.

The probability that this tablecloth has no faults is 0.0907

- (a) Find the value of A (3)

The tablecloths are sold in packets of 20

A randomly selected packet is taken.

- (b) Find the probability that more than 1 of the tablecloths in this packet has no faults. (3)

A hotel places an order for 100 tablecloths each of area A square metres.

The random variable X represents the number of these tablecloths that have no faults.

- (c) Find
- (i) $E(X)$
- (ii) $\text{Var}(X)$ (3)

- (d) Use a Poisson approximation to estimate $P(X = 10)$ (2)

It is claimed that a new machine produces cloth with a rate of faults that is less than 0.4 per square metre.

A piece of cloth produced by this new machine is taken at random.

The piece of cloth has area 30 square metres and is found to have 6 faults.

- (e) Stating your hypotheses clearly, use a suitable test to assess the claim made for the new machine. Use a 5% level of significance. (4)

- (f) Write down the p -value for the test used in part (e). (1)



4. Table 1 below shows the number of car breakdowns in the *Snoreap* district in each of 60 months.

Number of car breakdowns	0	1	2	3	4	5
Frequency	12	11	19	14	3	1

Table 1

Anja believes that the number of car breakdowns per month in *Snoreap* can be modelled by a Poisson distribution. Table 2 below shows the results of some of her calculations.

Number of car breakdowns	0	1	2	3	4	≥ 5
Observed frequency (O_i)	12	11	19	14	3	1
Expected frequency (E_i)	9.92			9.64	4.34	

Table 2

- (a) State suitable hypotheses for a test to investigate Anja's belief. (1)
- (b) Explain why Anja has changed the label of the final column to ≥ 5 (1)
- (c) Showing your working clearly, complete Table 2 (4)
- (d) Find the value of $\frac{(O_i - E_i)^2}{E_i}$ when the number of car breakdowns is
- (i) 1
- (ii) 3 (2)
- (e) Explain why Anja used 3 degrees of freedom for her test. (2)

The test statistic for Anja's test is 6.54 to 2 decimal places.

- (f) Stating the critical value and using a 5% level of significance, complete Anja's test. (2)



2. A manager keeps a record of accidents in a canteen.

Accidents occur randomly with an average of 2.7 per month. The manager decides to model the number of accidents with a Poisson distribution.

- (a) Give a reason why a Poisson distribution could be a suitable model in this situation. (1)
- (b) Assuming that a Poisson model is suitable, find the probability of
- (i) at least 3 accidents in the next month, (1)
- (ii) no more than 10 accidents in a 3-month period, (2)
- (iii) at least 2 months with no accidents in an 8-month period. (4)

One day, two members of staff bump into each other in the canteen and each report the accident to the manager. The canteen manager is unsure whether to record this as one or two accidents.

Given that the manager still wants to model the number of accidents per month with a Poisson distribution,

- (c) state
- a property of the Poisson distribution that the manager should consider when deciding how to record this situation
 - whether the manager should record this as one or two accidents
- (1)

The manager introduces some new procedures to try and reduce the average number of accidents per month.

During the following 12 months the total number of accidents is 22
The manager claims that the accident rate has been reduced.

- (d) Use a 5% level of significance to carry out a suitable test to assess the manager's claim.
You should state your hypotheses clearly and the p -value used in your test. (4)



3. Two car hire companies hire cars independently of each other.

Car Hire *A* hires cars at a rate of 2.6 cars per hour.

Car Hire *B* hires cars at a rate of 1.2 cars per hour.

- (a) In a 1 hour period, find the probability that each company hires exactly 2 cars. (2)

- (b) In a 1 hour period, find the probability that the total number of cars hired by the two companies is 3 (2)

- (c) In a 2 hour period, find the probability that the total number of cars hired by the two companies is less than 9 (2)

On average, 1 in 250 new cars produced at a factory has a defect.

In a random sample of 600 new cars produced at the factory,

- (d) (i) find the mean of the number of cars with a defect,
(ii) find the variance of the number of cars with a defect. (2)

- (e) (i) Use a Poisson approximation to find the probability that no more than 4 of the cars in the sample have a defect.
(ii) Give a reason to support the use of a Poisson approximation. (2)

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- (2)

(1)

(2)

(6)